

Explain why a computer system requires an Operating System (OS)

9608_w19_qp_12_sc_1.a

QUESTION

- 1 (a) The diagram shows different types of software on the left, and descriptions on the right.

Draw a line from each type of software to its correct description.

Type of software

Operating system

Utility program

Library program

Compiler

Description

Provides a ready-built routine that can be imported into a program

Provides an interface between the user and the hardware

Converts source code into a low-level language

Creates a new document for the user to edit

An additional program that helps to maintain or configure the system

[4]

MARK SCHEME

1(a)

1 mark per correct line

4

Type of software

Description

Operating
system

Utility
program

Library
program

Compiler

Provides a ready-built routine that
can be imported into a program

Provides an interface between
the user and the hardware

Converts source code into a
low-level language

Creates a new document for
the user to edit

An additional program that helps
to maintain or configure the
system

QUESTION

- (b) Aaron's computer has an operating system (OS). The OS manages the running processes and provides a user interface.

Describe these OS management tasks.

Process management

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Provision of a user interface

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[6]

MARK SCHEME

2(b)	1 mark per bullet point to max 4 for each management task, max 6 in total Process management: - Manages the scheduling of processes ... allows multi-tasking / multi-processing ... ensures fair access ... handles priorities - Manages the resources the processes need - Enables processes to share information - Prevents interference between processes// resolution of conflicts Provision of a user interface: - Allows a user to communicate with the hardware // vice-versa ... by making navigation around the system easier - Provides facility for user inputting data - Provides facility for outputting to the user - By example e.g. command line / GUI / menu-driven	6
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QUESTION

- 4 (a) (i) Explain why a personal computer (PC) needs an operating system (OS).

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[2]

MARK SCHEME

4(a)(i)	Two from: The hardware is unusable without an OS // hides complexity of hardware from user 1 Acts as an interface/ controls communications between user and hardware / hardware and software // or by example 1 Provides software <u>platform / environment</u> on which other programs can be run 1	2
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QUESTION

- (b) (i) Explain why a computer needs an operating system.

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[2]

MARK SCHEME

(b) (i) Any two from - The hardware is unusable without an OS // hides complexity of hardware from user - Acts as an interface / controls communications between user and hardware / hardware and software - Provides software platform / environment on which other programs can be run	[2]
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QUESTION

(b) (i) Explain why a computer needs an operating system.

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.....[2]

MARK SCHEME

(b) (i) Any **two** from

- The hardware is unusable without an OS // hides complexity of hardware from user
- Acts as an interface / controls communications between user and hardware / hardware and software
- Provides software platform / environment on which other programs can be run [2]

Explain the key management tasks carried out by the Operating System

9608_w21_qp_11_sc_7.a.i

QUESTION

7 (a) A computer has an operating system (OS).

(i) State **three** memory management tasks performed by the OS.

- 1
- 2
- 3
- [3]

MARK SCHEME

7(a)(i)	One mark per bullet point to max 3 • Prevents two processes occupying the same memory space • Organises memory e.g. paging/segmentation • Makes uses of virtual memory • Keeps track of allocated and free memory locations • Allocates memory to processes • Ensures fair use of memory • Releases memory when a process stops	3
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9608_w21_qp_11_sc_7.a.ii

QUESTION

(ii) State **two** input/output device management tasks performed by the OS.

- 1
- 2
- [2]

MARK SCHEME

7(a)(ii)	One mark per bullet point to max 2 • Installs drivers • Sends/receives data from a buffer // buffer management // • Sends commands to the device • Receives/handles messages/signals/interrupts from the device • Control of hardware usage by processes • Device detection • Power management • Keep track of device status (free or busy)	2
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9608_s21_qp_13_sc_4.a.i

QUESTION

4 An operating system (OS) is installed on a computer.

(a) The OS performs a number of different tasks such as device management and error detection and recovery.

(i) State **three** device management tasks the OS performs.

1

2

3 [3]

MARK SCHEME

4(a)(i)	1 mark per bullet point to max 3 • Install/manage device drivers • Control of hardware usage by processes // allocation of devices to processes // inter process communication • Device detection • Power Management • Keep track of device status (free or busy) • Buffer management	3
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9608_s21_qp_13_sc_4.a.ii

QUESTION

(ii) State **three** error detection and recovery management tasks the OS performs.

1

2

3 [3]

MARK SCHEME

4(a)(ii)	1 mark per bullet point to max 3 • Deals with interrupts • Deal with run time errors generated by software • Deal with hardware faults • Error diagnostic messages • Deadlock detection and recovery • Safe-mode boot-up routines • System shutdown • Saves system restore points	3
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9608_s21_qp_13_sc_4.a.iii

QUESTION

(iii) State **two** tasks, other than device management and error detection and recovery management that are carried out by an OS.

1

2 [2]

MARK SCHEME

4(a)(iii)	1 mark per bullet point to max 2 • Process / task/resource management • Main memory management • File/secondary storage management • Security management • Provision of a software platform/environment on which other programs can be run • Interrupt handling • Provision of a user interface NOT • Peripheral/hardware/device management • Error detection and recovery management	2
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9608_s20_qp_12_sc_2.c.ii

QUESTION

(ii) Amir's computer has several peripheral devices connected to it.

State **three peripheral** management tasks performed by the operating system.

Task 1

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Task 2

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Task 3

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[3]

MARK SCHEME

2(c)(ii)	1 mark per bullet point to max 3 • Installation of device driver software • Managing interrupts / signals from the device • Sending control signals to the device • Control of buffers • Management of queues	3
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QUESTION

- 8 Bart plays computer games on his stand-alone games console.

The games console has an operating system.

- (a) Describe the tasks performed by the operating system to manage the **main memory** in the games console.

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..... [4]

MARK SCHEME

8(a)	1 mark per bullet point to max 4 • Reads/writes data to/from RAM • ... e.g. current data/instructions from a game so the CPU can access it • Allocates virtual memory • ... when there is insufficient RAM to run a program/game • Allocates RAM to optimise performance • Paging • Segmentation	4
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QUESTION

2 Leonardo's mobile phone has an operating system (OS).

(a) Describe the following key management tasks that the mobile phone operating system carries out.

Process management

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Memory management

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[6]

MARK SCHEME

2(a)	1 mark per bullet point to max 4 for each management task, max 6 in total Process Management Manages the scheduling of processes ... allows multi-tasking / multi-processing ... ensures fair access ... handles priorities Manages which resources the processes require Enables processes to share information Prevents interference between processes // resolution of conflicts Memory Management Allocates memory to processes Ensures fair usage of memory Organises memory / by example Makes use of virtual memory Keep processes separate To release memory when a process stops	6
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QUESTION

- (b) Aaron's computer has an operating system (OS). The OS manages the running processes and provides a user interface.

Describe these OS management tasks.

Process management

.....

.....

.....

.....

.....

Provision of a user interface

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.....

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[6]

MARK SCHEME

2(b)	1 mark per bullet point to max 4 for each management task, max 6 in total Process management: - Manages the scheduling of processes ... allows multi-tasking / multi-processing ... ensures fair access ... handles priorities Manages the resources the processes need Enables processes to share information Prevents interference between processes// resolution of conflicts Provision of a user interface: - Allows a user to communicate with the hardware // vice-versa ... by making navigation around the system easier - Provides facility for user inputting data - Provides facility for outputting to the user - By example e.g. command line / GUI / menu-driven	6
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QUESTION

- 1 A computer has an operating system (OS) and utility software.

- (a) The following table lists key management tasks performed by an operating system and their descriptions.

Complete the table by writing the missing management task names and descriptions.

Management task	Description
Memory management	
	Provides user accounts and passwords
	Handles the signals sent when the attention of the processor is required elsewhere
Provision of a software platform	

[4]

MARK SCHEME

1(a)	1 mark for each correct entry	4										
	<table><tr><th>Management task</th><th>Description</th></tr><tr><td>Memory management</td><td>Handles the allocation of memory to processes // Ensures two programs do not attempt to use the same memory locations // Keeps track of allocated and free memory locations</td></tr><tr><td>Security management</td><td>Provides user accounts and passwords</td></tr><tr><td>Interrupt processing</td><td>Handles the signals sent when the attention of the processor is required elsewhere</td></tr><tr><td>Provision of a software platform</td><td>Provides an environment within which programs can be run</td></tr></table>	Management task	Description	Memory management	Handles the allocation of memory to processes // Ensures two programs do not attempt to use the same memory locations // Keeps track of allocated and free memory locations	Security management	Provides user accounts and passwords	Interrupt processing	Handles the signals sent when the attention of the processor is required elsewhere	Provision of a software platform	Provides an environment within which programs can be run	
Management task	Description											
Memory management	Handles the allocation of memory to processes // Ensures two programs do not attempt to use the same memory locations // Keeps track of allocated and free memory locations											
Security management	Provides user accounts and passwords											
Interrupt processing	Handles the signals sent when the attention of the processor is required elsewhere											
Provision of a software platform	Provides an environment within which programs can be run											

QUESTION

- 1 An operating system (OS) is usually pre-installed on a new computer.
- (a) The OS performs a number of different tasks such as file management and peripheral management.
- (i) State **three** file management tasks the OS performs.
- 1
- 2
- 3
- [3]

MARK SCHEME

1(a)(i)	1 mark per bullet to max 3 Storage space divided into file allocation units Space allocated to particular files Maintains/creates directory structures Specifies the logical method of file storage (e.g. FAT or NTFS) Provides file naming conventions Controls access // implements access rights // implements password protection // Makes file sharing possible Specifies tasks that can be performed on a file (e.g. open, close, delete, copy, create, move etc.)	3
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QUESTION

- (ii) State **three** printer management tasks the OS performs.
- 1
- 2
- 3
- [3]

MARK SCHEME

1(a)(ii)	1 mark per bullet to max 3 Installs printer driver Sends data to the printer / buffer to print // sends documents to the print queue Sends commands to printer Receives and handles (error) messages/signals/interrupts from the printer	3
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9608_s18_qp_11_sc_3.a.i

QUESTION

3 An operating system (OS) is usually pre-installed on a new computer.

(a) The OS performs a number of different tasks such as memory management and security management.

(i) State **three** memory management tasks the OS performs.

1

2

3

[3]

MARK SCHEME

3(a)(i)	1 mark per bullet to max 3 Allocates / deallocates RAM to programs/tasks/processes Keeps track of allocated and free memory locations Swaps data to and from the hard drive Handles virtual memory Paging // segmentation Memory protection, preventing a process accessing memory not allocated to it	3
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9608_s18_qp_11_sc_3.a.ii

QUESTION

(ii) State **three** security management tasks the OS performs.

- 1
- 2
- 3
- [3]

MARK SCHEME

3(a)(ii)	1 mark per bullet to max 3 Sets up user accounts Checks usernames, passwords // Authentication Implements access rights <u>Automatic</u> backup System restore / roll back (to previous stable state)	3
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9608_s18_qp_11_sc_3.a.iii

QUESTION

(iii) State **two** tasks, other than memory management and security management that are carried out by an OS.

- 1
- 2
- [2]

MARK SCHEME

3(a)(iii)	1 mark per bullet to max 2 Device / peripheral management File management Process management Input / output management Error detection / recovery Provides a user interface Facilitates communication between hardware and software / hardware devices	2
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9608_w17_qp_12_sc_1

QUESTION

- 1 One of the tasks of the operating system (OS) is the management of the main memory of the computer system.

State and describe **three** other operating system management tasks.

- 1
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-
- 2
-
-
- 3
-
-

[6]

MARK SCHEME

1	1 Mark for stating the management task 1 Mark for a corresponding description Maximum 2 marks for each task Maximum 3 tasks Process / Task Management Allocation of processor time Scheduling of processes or tasks / multi-tasking / multi-programming etc. By example – e.g. round-robin, shortest remaining time first etc. Resolution of conflict when two or more processes require the same resource Secondary Storage management Storage space divided into file allocation units Space allocated to particular files OS maintains a file directory and FAT Provides file naming conventions Controls access. Peripheral / Hardware / Device / Input/output Management Installation of appropriate driver software Controls access to data being sent to/from hardware/peripherals Controls access to hardware/peripherals Manages communication between devices / hardware and software Provision of a User interface Allows user interaction with the computer system// Facilitates human computer communication Hides the complexity of the hardware from the user Or by example – e.g. GUI, command line etc. Interrupt Handling Halts the execution of the current process Stores the values of the current process on the stack Loads and executes the appropriate ISR code Use of priorities for handling simultaneous interrupts Saves data on power outage Security Management Makes provision for recovery when data is lost Provides usernames and passwords / encryption / user accounts Prevents unauthorised access Ensures privacy of data Provision of a software platform / environment On which other programs / applications can be run	6
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QUESTION

- 6 (a) The operating system (OS) contains code for performing various management tasks.

The appropriate code is run when the user performs various actions.

Draw a line to link each OS management task to the appropriate user action.

OS management task	Action
Main memory management	The user moves the mouse on the desktop
Input/Output management	The user closes the spreadsheet program
Secondary storage management	The user selects the Save command to save their spreadsheet file
Human computer interface management	The user selects the Print command to output their spreadsheet document

[3]

MARK SCHEME

6(a)	One mark for each correct line from each left hand box to max three marks.	3
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QUESTION

- (ii) One of the tasks carried out by the OS is the management of the use of the processor.

Name and describe **two** other management tasks that the OS performs.

1

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2

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.....

[4]

MARK SCHEME

4(a)(ii)	One mark for the name and one mark for description. Max two management tasks.	Max 4
	Provides the Human Computer Interface (HCI)	1
	Controls communications between user and hardware// or by example	1
	Main memory management	1
	Memory protection to ensure that two programs do not try to use the same space // Use of virtual memory // Location of processes within the memory // By example 1	
	File / Secondary storage management	1
	Maintains directory structures // Provides file naming conventions // Controls access	1
	Peripheral / hardware / device / Input-Output management	1
	Installation of appropriate driver software // Controls access to data being sent to/from hardware/peripherals // Controls access to hardware/peripherals // manages communication between devices.	1
	Interrupt handling	1
	Identifies priorities of interrupts // Saves data on power outage // Loads appropriate Interrupt Service Routine (ISR) // By example	1
	Security management	1
	Makes provision for recovery when data is lost // Provides usernames and passwords // Prevents unauthorised access // Ensures privacy of data	1

QUESTION

- 6 (a) The operating system (OS) contains code for performing various management tasks.

The appropriate code is run when the user performs various actions.

Draw a line to link each OS management task to the appropriate user action.

OS management task	Action
Main memory management	The user moves the mouse on the desktop
Input/Output management	The user closes the spreadsheet program
Secondary storage management	The user selects the Save command to save their spreadsheet file
Human computer interface management	The user selects the Print command to output their spreadsheet document

[3]

MARK SCHEME

6(a)	One mark for each correct line from each left hand box to max three marks.	3
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QUESTION

(ii) Give **two** key management tasks carried out by an operating system.

- 1
.....
2
.....[2]

MARK SCHEME

(ii) Any **two** from:

- Process / task / resource management
- Main memory management
- Peripheral / hardware / device management
- File / secondary storage management
- Security management
- Provision of a software platform / environment on which other programs can be run
– only if not given in part (b)(i)
- Interrupt handling
- Provision of a user interface run – only if not given in part (b)(i) [2]

QUESTION

- 3 When an application program requests a file stored on a hard disk, the computer system reads the file. Use the statement labels A to H to complete the sequence of steps that describe how this happens.

Label	Statement
A	When the hard disk drive has read the file, it generates an interrupt.
B	While the file continues, the head reads successive clusters of sectors from the disk and writes data into the disk buffer.
C	The head reads the first cluster of sectors from disk and writes data into the disk buffer.
D	The head moves to the correct track.
E	The operating system transfers the contents of the disk buffer to the application program's data memory.
F	In the relevant directory file, the operating system looks up the track and sector where the file begins.
G	Application program passes file read request to the operating system.
H	The hard disk drive waits for the correct sector to arrive under the head.

1. The application program executes a statement to read a file.
2.
3. The operating system begins to spin the hard disk, if it is not currently spinning.
4.
5.
6.
7.
8.
9.
10.

[8]

MARK SCHEME

3 **ONE** mark for each letter in the correct place.

Then **ONE** mark for any pair of letters in the correct order, but not in the correct place

- | | |
|----|---|
| 1 | The application program executes a statement to read a file. |
| 2 | G |
| 3 | The operating system begins to spin the hard disk, if it is not currently spinning. |
| 4 | F |
| 5 | D |
| 6 | H |
| 7 | C |
| 8 | B |
| 9 | A |
| 10 | E |

[8]

9608_w16_qp_12_sc_7

QUESTION

7 One management task carried out by an operating system is to provide a user interface.

Describe **two** more of these management tasks.

- 1
-
-
-
- 2
-
-
-
-[4]

MARK SCHEME

7 **ONE** mark per bullet point, **MAX TWO** marks per task.

- Process /resource management
- Scheduling of processes /multi-tasking /multi-programming etc.
- Resolution of conflicts when two or more processes require the same resource
- Main memory management
- Memory protection to ensure that two programs do not try to use the same space
- Use of virtual memory
- Deciding which processes need to be in main memory at any one time
- Location of processes within the memory
- By example, e.g. when process terminates, memory is made available
- Peripheral /hardware /device management
- Installation of appropriate driver software
- Controls access to data being sent to /from hardware /peripherals
- Controls access to hardware /peripherals
- Manages communication between devices /hardware and software
- File /secondary storage management
- Maintains directory structures
- Provides file naming conventions
- Controls access
- Security management
- Makes provision for recovery when data is lost
- Provides usernames and passwords /encryption /user accounts
- Prevents unauthorised access
- Ensures privacy of data
- Provision of a software platform /environment
- On which other programs can be run
- Interrupt handling
- Identifies priorities of interrupts
- Save current memory /process values /saves data on power outage
- Loads appropriate Interrupt Service Routine (ISR)
- Any relevant example

[4]

QUESTION

(ii) Give **two** key management tasks carried out by an operating system.

- 1
.....
- 2
.....[2]

MARK SCHEME

(ii) Any **two** from:

- Process / task / resource management
- Main memory management
- Peripheral / hardware / device management
- File / secondary storage management
- Security management
- Provision of a software platform / environment on which other programs can be run
– only if not given in part (b)(i)
- Interrupt handling
- Provision of a user interface run – only if not given in part (b)(i) [2]

Show understanding of the need for typical utility software provided with an Operating System

9608_w21_qp_11_sc_7.b

QUESTION

(b) Utility programs are also provided with the OS.

Identify **and** describe **two** different utility programs.

Utility program

Description

.....

Utility program

Description

.....

[4]

MARK SCHEME

7(b)	One mark for name and one for description × 2, max 4 The description must match the name • Virus/Malware checker • ...scans disk for viruses and reports, quarantines or deletes them • File compression • ...reduces the size of a file • Back up • ...creates a copy of data in case of loss, so that the images can be restored • Defragmenter • ...finds files that are not stored in contiguous blocks and moves them together • Disk formatter • ...creates a logical drive on a hard drive // reformats a previously used hard drive // creates logical partitions // prepares a disk for first use • Disk repair/Disk contents analysis • ...attempts to recover damaged files // checks disk for space and usage // to identify/mark the errors / bad sectors	4
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QUESTION

(b) Utility programs are installed on a new computer.

(i) The following table lists six programs.

Tick (✓) **one** box in each row to identify whether the program is a utility program or not a utility program.

Program	Utility	Not utility
Language translator		
Backup		
Integrated Development Environment (IDE)		
Graphics		
Defragmenter		
Spreadsheet		

[2]

MARK SCHEME

4(b)(i)	1 mark for first 3 rows 1 mark for last 3 rows	2																					
	<table> <tr> <th>Program</th><th>Utility</th><th>Not utility</th></tr> <tr> <td>Language translator</td><td></td><td>✓</td></tr> <tr> <td>Backup</td><td>✓</td><td></td></tr> <tr> <td>Integrated Development Environment (IDE)</td><td></td><td>✓</td></tr> <tr> <td>Graphics</td><td></td><td>✓</td></tr> <tr> <td>Defragmenter</td><td>✓</td><td></td></tr> <tr> <td>Spreadsheet</td><td></td><td>✓</td></tr> </table>	Program	Utility	Not utility	Language translator		✓	Backup	✓		Integrated Development Environment (IDE)		✓	Graphics		✓	Defragmenter	✓		Spreadsheet		✓	
Program	Utility	Not utility																					
Language translator		✓																					
Backup	✓																						
Integrated Development Environment (IDE)		✓																					
Graphics		✓																					
Defragmenter	✓																						
Spreadsheet		✓																					

9608_s21_qp_13_sc_4.b.ii

QUESTION

(ii) Identify **two other** utility programs.

1

2 [2]

MARK SCHEME

4(b)(ii)	1 mark per bullet point to max 2 e.g. • Virus checker // anti-malware • Disk formatter • Disk contents analysis/disk repair • System clean-up • File compression • Firewall • Encryption	2
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9608_s21_qp_12_sc_7.a.i

QUESTION

7 Sam is a photographer. She has an image library of over 10 000 images. She stores the images on a high capacity magnetic hard disk.

(a) Explain why Sam would use the following utility software.

(i) Backup

.....

.....

.....

..... [2]

MARK SCHEME

7(a)(i)	1 mark per bullet point to max 2 • The images on the hard disk can be lost or corrupted • To make a copy of her files (which can be stored elsewhere) • So that the images can be restored	2
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9608_s21_qp_12_sc_7.a.ii

QUESTION

(ii) Defragmenter

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.....
.....
..... [2]

MARK SCHEME

7(a)(ii)	1 mark per bullet point to max 2 • Frequent changes to the images mean the data for each file is split across the disk • To rearrange the (fragmented) files into contiguous locations // to bring all the empty space together • To improve the time it takes to access and load the files	2
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9608_s21_qp_12_sc_7.a.iii

QUESTION

(iii) Disk repair

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..... [2]

MARK SCHEME

7(a)(iii)	1 mark per bullet point to max 2 • Areas of the disk can become corrupt • To identify/mark the errors/bad sectors • ...so they can be repaired // so bad sectors are no longer used	2
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QUESTION

(c) Amir's computer has system software, including utility software and an operating system.

- (i) Explain how the disk formatter, disk contents analysis and disk repair utilities work together.

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..... [3]

MARK SCHEME

2(c)(i)	1 mark per bullet point to max 3 • Disk contents analysis checks for errors/problems with the disk • Disk repair attempts to fix the errors • The disk formatter prepares the disk for (initial) use (again).	3
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QUESTION

9 Utility programs are examples of system software.

(a) Complete the table by writing the name of the utility program for each description.

Description	Utility program
Reorganises files on a disk to improve efficiency	
Scans a hard disk to identify bad sectors	
Prepares a hard disk for first use	

[3]

MARK SCHEME

9(a)	1 mark for each correctly identified utility program		3
	Description	Utility program	
	Reorganises files on a disk to improve efficiency	Defragmentation software	
	Scans a hard disk to identify bad sectors	Disk contents analysis / repair software	
	Prepares a hard disk for first use	Disk formatter	

9608_w19_qp_12_sc_1.a

QUESTION

- 1 (a) The diagram shows different types of software on the left, and descriptions on the right.

Draw a line from each type of software to its correct description.

Type of software

Operating system

Utility program

Library program

Compiler

Description

Provides a ready-built routine that can be imported into a program

Provides an interface between the user and the hardware

Converts source code into a low-level language

Creates a new document for the user to edit

An additional program that helps to maintain or configure the system

[4]

MARK SCHEME

1(a)	1 mark per correct line	4												
<table><thead><tr><th>Type of software</th><th>Description</th></tr></thead><tbody><tr><td>Operating system</td><td>Provides a ready-built routine that can be imported into a program</td></tr><tr><td>Utility program</td><td>Provides an interface between the user and the hardware</td></tr><tr><td>Library program</td><td>Converts source code into a low-level language</td></tr><tr><td>Compiler</td><td>Creates a new document for the user to edit</td></tr><tr><td></td><td>An additional program that helps to maintain or configure the system</td></tr></tbody></table>			Type of software	Description	Operating system	Provides a ready-built routine that can be imported into a program	Utility program	Provides an interface between the user and the hardware	Library program	Converts source code into a low-level language	Compiler	Creates a new document for the user to edit		An additional program that helps to maintain or configure the system
Type of software	Description													
Operating system	Provides a ready-built routine that can be imported into a program													
Utility program	Provides an interface between the user and the hardware													
Library program	Converts source code into a low-level language													
Compiler	Creates a new document for the user to edit													
	An additional program that helps to maintain or configure the system													

9608_w19_qp_12_sc_1.b

QUESTION

(b) Describe the purpose of disk repair software.

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..... [3]

MARK SCHEME

1(b)	1 mark per bullet point to max 3 Disk repair: - Checks for any errors / inconsistencies / bad sectors on the disk - Resolves any errors on the disk - Retrieves files / data from a damaged disk // re-constructs directory // - recovers disc when data corrupt - Marks bad sectors on the disk // marks bad sectors as unusable	3
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QUESTION

- (c) Aaron's computer has a virus checker and backup software.

Describe these utility programs.

Virus checker

.....

.....

.....

Backup software

.....

.....

.....

[4]

MARK SCHEME

2(c)	1 mark per bullet point to max 3 for each utility program, max 4 in total Virus checker: Scans files stored on a computer system for malicious code - Scans files when they enter the system / memory stick inserted / download etc. - Sets up a schedule for virus-checking - Isolates / quarantines / deletes viruses - Regularly updates the virus definitions Backup software: - Creates a copy of the contents of a disk / partition. Can be set up to automatically backup // schedules backups - Allows the user to decide what is backed up, e.g. all data // all files that have changed since the last backup - Allows the user to set up an off-site backup - May encrypt the backup files - Restores the data if necessary	4
------	---	----------

QUESTION

(b) A hard disk formatter and a hard disk defragmenter are two examples of utility software.

(i) Describe the actions performed by a hard disk formatter and a hard disk defragmenter.

Hard disk formatter

.....

.....

.....

Hard disk defragmenter

.....

.....

.....

[4]

MARK SCHEME

1(b)(i)	1 mark per bullet point to max 2 for formatter, max 2 for defragmenter hard disk formatter - Makes existing data inaccessible - Partitions the disk into logical drives - Sets up the (specified) file system - Prepares the disk for initial use - May check for errors on the disk hard disk defragmenter - Re-organises the disk contents - Moves split files so they are contiguous - Creates a larger area of (contiguous) free space	4
---------	--	----------

QUESTION

(ii) Identify **three other** examples of utility software that can be installed on the computer.

- 1
- 2
- 3

[3]

MARK SCHEME

1(b)(ii)	1 mark per bullet point For example: Backup software File compression Virus checker Disk contents analysis / repair	3
----------	---	----------

QUESTION

(b) Utility software is usually pre-installed on a new computer.

(i) The following table lists four programs. Put **one** tick (✓) in each row to indicate whether or not the program is utility software.

Program	True	False
Database		
Virus checker		
Web browser		
Backup software		

[4]

MARK SCHEME

1(b)(i)	1 mark for each correct box ticked.	4															
	<table><tr><th>Program</th><th>True</th><th>False</th></tr><tr><td>Database</td><td></td><td>✓</td></tr><tr><td>Virus checker</td><td>✓</td><td></td></tr><tr><td>Web browser</td><td></td><td>✓</td></tr><tr><td>Backup software</td><td>✓</td><td></td></tr></table>	Program	True	False	Database		✓	Virus checker	✓		Web browser		✓	Backup software	✓		
Program	True	False															
Database		✓															
Virus checker	✓																
Web browser		✓															
Backup software	✓																

9608_s18_qp_12_sc_1.b.ii

QUESTION

(ii) Name **two** other utility programs.

Program 1

Program 2 [2]

MARK SCHEME

1(b)(ii)	1 mark for each valid utility program to max 2 e.g. System clean up Automatic update Disk contents analysis / Disk checking / Disk repair File compression Disk formatter Firewall Disk Defragmenter	2
----------	--	----------

QUESTION

- (b) Utility software is usually pre-installed on a new computer.

The following table lists four programs. Put **one** tick (✓) in each row to indicate whether or not the program is utility software.

Program	True	False
Disk Defragmenter		
Word Processor		
Library Program		
Compression Software		

[4]

MARK SCHEME

3(b)	1 mark for each correct box ticked	4															
<table> <tr> <th>Programs</th><th>True</th><th>False</th></tr> <tr> <td>Disk Defragmenter</td><td>✓</td><td></td></tr> <tr> <td>Word Processor</td><td></td><td>✓</td></tr> <tr> <td>Library program</td><td></td><td>✓</td></tr> <tr> <td>Compression Software</td><td>✓</td><td></td></tr> </table>			Programs	True	False	Disk Defragmenter	✓		Word Processor		✓	Library program		✓	Compression Software	✓	
Programs	True	False															
Disk Defragmenter	✓																
Word Processor		✓															
Library program		✓															
Compression Software	✓																

QUESTION

- (b) A user has the following issues with the use of his PC.

State the utility software which should provide a solution.

- (i) The hard disk stores a large number of video files. The computer frequently runs out of storage space.

Utility software solution[1]

MARK SCHEME

6(b)(i)	File compression software	1
---------	---------------------------	---

9608_s17_qp_13_sc_6.b.ii

QUESTION

- (ii) The user is unable to find an important document. He thinks it was deleted in error some weeks ago. This must not happen again.

Utility software solution[1]

MARK SCHEME

6(b)(ii)	Backup software	1
----------	-----------------	---

9608_s17_qp_13_sc_6.b.iii

QUESTION

- (iii) The operating system reports 'Bad sector' errors.

Utility software solution[1]

MARK SCHEME

6(b)(iii)	Disk repair software	1
-----------	----------------------	---

9608_s17_qp_13_sc_6.b.iv

QUESTION

- (iv) There have been some unexplained images and advertisements appearing on the screen. The user suspects it is malware.

Utility software solution[1]

MARK SCHEME

6(b)(iv)	Anti-virus software	1
----------	---------------------	---

9608_s17_qp_12_sc_4.b.i

QUESTION

- (b) A user has the following issues with the use of their personal computer (PC).

For each case, state the utility software which should provide a solution.

- (i) The user wants to send a large file as an attachment to an email. The user knows that the recipient's Internet Service Provider (ISP) has a limit of 2MB for file attachments.

Utility software solution:[1]

MARK SCHEME

4(b)(i)	File compression software	1
---------	---------------------------	---

9608_s17_qp_12_sc_4.b.ii

QUESTION

- (ii) The user is writing a book and is worried that the document files could get damaged or deleted.

Utility software solution:[1]

MARK SCHEME

4(b)(ii)	Backup software	1
----------	-----------------	---

9608_s17_qp_12_sc_4.b.iii

QUESTION

- (iii) The computer has recently been slow to load large files. The user has deleted a large number of small files to try to solve the problem. A friend has advised that there is a procedure which should be regularly carried out to reorganise file storage on the hard disk.

Utility software solution:[1]

MARK SCHEME

4(b)(iii)	Disk defragmenting software	1
-----------	-----------------------------	---

9608_s17_qp_12_sc_4.b.iv

QUESTION

- (iv) The user clicked on an attachment in an unsolicited email. Since then, the computer has shown some unexplained behaviours.

Utility software solution:[1]

MARK SCHEME

4(b)(iv)	Anti-virus software	1
----------	---------------------	---

9608_s17_qp_11_sc_6.b.i

QUESTION

- (b) A user has the following issues with the use of his PC.

State the utility software which should provide a solution.

- (i) The hard disk stores a large number of video files. The computer frequently runs out of storage space.

Utility software solution[1]

MARK SCHEME

6(b)(i)	File compression software	1
---------	---------------------------	---

9608_s17_qp_11_sc_6.b.ii

QUESTION

- (ii) The user is unable to find an important document. He thinks it was deleted in error some weeks ago. This must not happen again.

Utility software solution[1]

MARK SCHEME

6(b)(ii)	Backup software	1
----------	-----------------	---

9608_s17_qp_11_sc_6.b.iii

QUESTION

(iii) The operating system reports 'Bad sector' errors.

Utility software solution[1]

MARK SCHEME

6(b)(iii)	Disk repair software	1
-----------	----------------------	---

9608_s17_qp_11_sc_6.b.iv

QUESTION

(iv) There have been some unexplained images and advertisements appearing on the screen. The user suspects it is malware.

Utility software solution[1]

MARK SCHEME

6(b)(iv)	Anti-virus software	1
----------	---------------------	---

QUESTION

- 7 A small company produces scientific magazines. The owner buys some new desktop computers. The computers are used to store thousands of colour images (diagrams and photographs). All the computers have Internet access.

(a) Name **three** utility programs the company would use on all their computers. Describe what each program does.

1

Description

.....

2

Description

.....

3

Description

.....[6]

MARK SCHEME

- 7 (a) **One mark** for the name and **one mark** for the explanation for **three** utility programs

- Disk formatter
- Prepares a hard disk to allow data to be stored on it
- Virus checker
- Checks for viruses and then quarantines removes any virus found
- File compression
- Reduces file size by removing redundant details (lossy / lossless)
- Backup software
- Makes copy of files on another medium in case of corruption / loss of data
- Firewall
- Prevents unauthorised access to computer system from external sources

[6]

QUESTION

- 7 A small company produces scientific magazines. The owner buys some new desktop computers. The computers are used to store thousands of colour images (diagrams and photographs). All the computers have Internet access.

(a) Name **three** utility programs the company would use on all their computers. Describe what each program does.

1

Description

.....

2

Description

.....

3

Description

.....[6]

MARK SCHEME

7 (a) **One mark** for the name and **one mark** for the explanation for **three** utility programs

- Disk formatter
- Prepares a hard disk to allow data to be stored on it
- Virus checker
- Checks for viruses and then quarantines removes any virus found
- File compression
- Reduces file size by removing redundant details (lossy / lossless)
- Backup software
- Makes copy of files on another medium in case of corruption / loss of data
- Firewall
- Prevents unauthorised access to computer system from external sources

[6]

9608_w15_qp_13_sc_10.b

QUESTION

(b) A virus checker has been installed on a PC.

Give **two** examples of when a virus checker should perform a check.

1

.....

2

.....

[2]

MARK SCHEME

(b) any **two** from:

- checks for boot sector viruses when machine is first turned on
- when an external storage device is connected
- checks a file / web page when it is accessed / downloaded

[2]

9608_w15_qp_11_sc_10.b

QUESTION

(b) A virus checker has been installed on a PC.

Give **two** examples of when a virus checker should perform a check.

1

.....

2

.....

[2]

MARK SCHEME

(b) any **two** from:

- checks for boot sector viruses when machine is first turned on
- when an external storage device is connected
- checks a file / web page when it is accessed / downloaded

[2]

QUESTION

(b) All of the uses in **part (a)** involve the input or output of data.

(i) Describe **two** methods of preventing accidental loss of data.

- 1
-
- 2
-[2]

MARK SCHEME

(b) (i) Any **two** from:

- frequent (or equivalent) backup EITHER to secondary media/to 3rd party server/cloud/removable devices/continuous backup OR stored remotely
- disk-mirroring strategy/RAID
- UPS (uninterruptable power supply)/backup generator

[2]

Show understanding of program libraries

9608_w21_qp_11_sc_7.c

QUESTION

(c) Some operating systems include library programs.

State **three** benefits to a programmer of using library programs.

- 1
.....
- 2
.....
- 3
.....
- [3]

MARK SCHEME

7(c)	One mark per bullet point to max 3 • Already tested • Should be relatively free from errors // more robust • Used by many others • Make use of another programmer's knowledge • Precompiled • Saves (programming/testing) time	3
------	--	----------

QUESTION

4 Annchi is writing a computer game with a group of friends.

(a) One of her friends has suggested using Dynamic Link Library (DLL) files to help them develop the game.

(i) Give **three** reasons why Annchi and her friends should use DLL files when developing the game.

1

.....

2

.....

3

.....

[3]

MARK SCHEME

4(a)(i)	1 mark per reason to max 3 • DDL file is only loaded into memory when required • ... so the executable file for the game is smaller • Changes/improvements in the DLL file are independent of the main program • ... the game program will not need to be recompiled • ... the game program will get the benefit of the updates automatically • The same DDL file can be used in several game programs (at the same time) • (DLL) routines are pre-written saving the developers time • (DLL) routines are pre-tested so should be reliable • Developers can take advantage of other programmers' expertise	3
---------	---	----------

QUESTION

(ii) Give **two** reasons why Annchi and her friends should **not** use DLL files when developing the game.

1

.....

2

.....

[2]

MARK SCHEME

4(a)(ii)	1 mark per reason to max 2 - Game will not work if DDL is corrupted - An external change to the DDL could stop the game working or change the way it works - The DDL file must be present at run-time otherwise there is an error	2
----------	--	----------

9608_w19_qp_13_sc_6.b

QUESTION

(b) Willow used functions from a JavaScript program library in the web page.

Describe the benefits to Willow of using program libraries to create the web page.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

6(b)	1 mark per bullet point to max 4 Using program libraries saves time ... as she does not have to write some routines Program library routines should already be thoroughly tested ... so, Willow can use them without having to spend time checking they work / correcting errors Program library routines may perform complex tasks ... so, Willow won't have to work out how to code them	4
------	---	----------

QUESTION

- 1 (a) The diagram shows different types of software on the left, and descriptions on the right.

Draw a line from each type of software to its correct description.

Type of software

Operating system

Utility program

Library program

Compiler

Description

Provides a ready-built routine that can be imported into a program

Provides an interface between the user and the hardware

Converts source code into a low-level language

Creates a new document for the user to edit

An additional program that helps to maintain or configure the system

[4]

MARK SCHEME

1(a)	1 mark per correct line	4												
<table><thead><tr><th>Type of software</th><th>Description</th></tr></thead><tbody><tr><td>Operating system</td><td>Provides a ready-built routine that can be imported into a program</td></tr><tr><td>Utility program</td><td>Provides an interface between the user and the hardware</td></tr><tr><td>Library program</td><td>Converts source code into a low-level language</td></tr><tr><td>Compiler</td><td>Creates a new document for the user to edit</td></tr><tr><td></td><td>An additional program that helps to maintain or configure the system</td></tr></tbody></table>		Type of software	Description	Operating system	Provides a ready-built routine that can be imported into a program	Utility program	Provides an interface between the user and the hardware	Library program	Converts source code into a low-level language	Compiler	Creates a new document for the user to edit		An additional program that helps to maintain or configure the system	
Type of software	Description													
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Utility program	Provides an interface between the user and the hardware													
Library program	Converts source code into a low-level language													
Compiler	Creates a new document for the user to edit													
	An additional program that helps to maintain or configure the system													

9608_s19_qp_12_sc_2.b

QUESTION

(b) Biyu uses library files in the program.

Explain why software is often developed using library files.

.....

.....

.....

..... [2]

MARK SCHEME

2(b)	1 mark per bullet point to max 2 • The code is already written so the programmer is not starting over again which saves time • The code will have been used by many people so it should be already thoroughly tested and relatively error-free / won't need re-testing • The programmer can use e.g. mathematical / graphics functions etc. that he may not know how to code • Use of named library functions can simplify the program and make it easier to read • The library routine code should conform to industry standards and therefore contribute towards a more robust program	2
------	--	----------

9608_s19_qp_11_sc_3.a.i

QUESTION

3 Kimmy has written a program in a high-level language.

(a) Kimmy has used library routines in the program.

(i) Describe **two** advantages of using library routines in the program.

- 1
-
-
-
- 2
-
-
-

[4]

MARK SCHEME

3(a)(i)	1 mark for each advantage, 1 mark for a valid expansion to max 2 · 2 Code is already tested ...so it is more robust/likely to work Saves programming time ...code does not have to be written/re-written from scratch The programmer can use e.g. mathematical functions ...that s/he may not know how to code If there is an improvement in the library routine ...the program updates automatically	4
---------	--	----------

9608_s19_qp_11_sc_3.a.ii

QUESTION

(ii) Describe what is meant by a **Dynamic Link Library** (DLL).

.....

.....

.....

..... [2]

MARK SCHEME

3(a)(ii)	1 mark per bullet point to max 2 A collection of self-contained (shared library) programs ...that are already compiled Linked to the main program during execution Library program code is separate from the .EXE file Library file only loaded into memory when required at run time A DLL file can be made available to several applications (at the same time) If DLL routine is updated the program that uses it will run the update	2
----------	--	----------

QUESTION

(b) Kim will use library routines in her program.

(i) Describe what is meant by a **library routine**.

.....

.....

.....

..... [2]

MARK SCHEME

6(b)(i)	1 mark per bullet to max 2 • Pre-existing / pre-compiled / pre-written modules / code ... • ... can be linked into her program (without amendment) • To perform common / complex tasks	2
---------	---	----------

QUESTION

(ii) Describe **one** benefit and **one** drawback of using library routines.

Benefit

.....

.....

.....

Drawback

.....

.....

..... [4]

MARK SCHEME

6(b)(ii)	1 mark per bullet point. Max 2 for <u>one benefit</u>, max 2 for <u>one drawback</u> Benefit: • Less code needs to be written • ... saves time / saves re-inventing the wheel • Pre-tested // Used by many people • ... reduces time testing // can be fairly sure that the function will perform as it should • Can be written in a different programming language • ... making use of special features of that language • Can be complex algorithms (e.g. mathematical/graphics functions) • ... she does not need to work out how to write it // ...that she may not know how to code • Simplifies the program • ... since just the name of the function included in the source code Drawback: • Compatibility issues • ... may not work with the other code/may require changing program for it to work • Not guaranteed thorough testing • ... may be unknown or unexpected bugs / virus • Library routine may not meet exact needs • ... may give unexpected results // ... may need editing • If library routine is changed • ... there may be unexpected results / errors	4
----------	---	----------

9608_w16_qp_13_sc_2.c

QUESTION

- (c) New program code is to be written in a high-level language. The use of Dynamic Link Library (DLL) files is considered in the design.

Describe what is meant by a DLL file.

.....

.....

.....

.....[2]

MARK SCHEME

(c) Any **two** from:

- A DLL file is a shared library file
- Code is saved separately from the main .EXE files
- Code is only loaded into main memory when required at run-time
- The DDL file can be made available to several applications (at the same time)

[2]

9608_w16_qp_12_sc_8.a

QUESTION

8 A programmer is writing a program that includes code from a program library.

(a) Describe **two** benefits to the programmer of using one or more library routines.

- 1
-
-
-
- 2
-
-
-
-[4]

MARK SCHEME

8 (a) ONE mark for each bullet point from **MAX TWO** groups.

- The code is already written
- (So the programmer is not starting over again) which saves time
- The code will have been used by many people
- So it should be already thoroughly tested // relatively error-free
- The programmer can use, e.g. mathematical / graphics functions, etc. (may not know how to code)
- Can be sure that the function will perform as it should // simplifies the program.
- The code should conform to industry standards
- And therefore contribute towards a more robust program

[4]

QUESTION

(b) The programmer decides to use a Dynamic Link Library (DLL) file.

(i) Describe **two** benefits of using DLL files.

- 1
-
-
-
- 2
-
-
-
-[4]

MARK SCHEME

(b) (i) **ONE** mark for each benefit, and **ONE** mark for a further expansion.

- The executable file is smaller / the executable does not contain all the library routines ...
- ... DLL files are only loaded into memory when required.
- Changes / improvements / error correction to the DLL file code are done independently of the main program...
- ... So there is no need to recompile the main program
- ... All programs using it will benefit
- A single DLL file can be made available to several application programs...
- ... Saving space in memory / easing the pressure on memory

[4]

QUESTION

(ii) State **one** drawback of using DLL files.

-
-
-[2]

MARK SCHEME

(ii) **ONE** mark for each bullet point from **MAX ONE** group.

- The executable code is not self-contained ...
- ... the DLL file(s) needed to be included at run time.
- Appropriate (linking) software must be available at run-time ...
- ... to link/include/import the DLL files.
- The DLL file must be present ...
- ... otherwise (unable to find X.dll) errors
- Unexpected changes to the DLL file/corrupted DLL file ...
- ... could mean the program stops working as expected
- Malicious changes to the DLL file ...
- ... could install a virus on the user's computer/related files could be corrupted

[2]

9608_w16_qp_11_sc_2.c

QUESTION

(c) New program code is to be written in a high-level language. The use of Dynamic Link Library (DLL) files is considered in the design.

Describe what is meant by a DLL file.

.....
.....
.....
.....[2]

MARK SCHEME

(c) Any **two** from:

- A DLL file is a shared library file
- Code is saved separately from the main .EXE files
- Code is only loaded into main memory when required at run-time
- The DDL file can be made available to several applications (at the same time)

[2]